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Travelling Salesperson Problem (TSP)

Exact Algorithms for TSP

Approximation Algorithm for TSP

Heuristics: A Brief Tour

Approximation Schemes and the Knapsack Problem

✓

 Videos: Fully Polynomial Time Approximation Scheme and Knapsack

29 min

📄

 Quiz: Fully Polynomial Time Approximation Scheme

30 min

Problem Set # 4

Fully Polynomial Time Approximation Scheme

Review Learning Objectives

Submit your assignment

Due Mar 3, 11:59 PM IST

Receive grade

To Pass 80% or higher

🎉 Congratulations! You passed!

Grade received 88.88%

Latest Submission Grade 88.89%

To pass 80% or higher

Go to next item

1. A fully polynomial time approximation scheme for a maximization problem guarantees a solution within $(1 - \epsilon)\text{OPT}$ where OPT is the optimal solution value in time $n^2 \times \left(\frac{1}{\epsilon}\right)^2$. We wish to compute an approximation that is guaranteed to be no worse than 0.9OPT . What is the time cost for the algorithm? 1 / 1 point
- ☐ Cannot say using the data given to us in this problem.

☒ $100n^2$

☐ $10n^2$

☐ n

☒ Correct

Try again

Your grade

88.88%

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The questions below concern a Knapsack problem instance with 4 items and the following data about weights and value.

	a1	a2	a3	a4
Weight	2	4	3	1
Value	1	2	1	1

The total weight budget is 565.

Recall that $W(i, v)$ is defined as the **minimum** weight of items needed to achieve value exactly v while choosing the subset of items $\{a_1, \dots, a_i\}$

Also recall the recurrence/base cases for W

$$W(0, 0) = 0, W(0, v) = \infty \text{ if } v > 0, W(i, 0) = 0$$

$$W(i, v) = \min(W(i - 1, v), W(i - 1, v - w_i) + w_i) \text{ where } w_i, v_i \text{ are the weight and value, respectively of item } a_i.$$

2. Which of the following expressions in terms of W yield the optimal value for the original Knapsack problem? 4 / 4 points
- ☐ $\arg \max_{v \leq 565} (W(3, v))$. Note that the $\arg \max$ yields the actual value v of the maximum.

☒ $\max\{v \mid 0 \leq v \leq 5 \text{ and } W(4, v) \leq 6\}$.

☐ $\min(W(4, 0), W(4, 1), W(4, 2), W(4, 3), W(4, 4), W(4, 5))$

☐ $\max\{v \mid 0 \leq v \leq 3 \text{ and } W(4, v) \leq 6\}$

☒ Correct

Correct. This yields the maximum value v such that the weight needed to achieve v comes in under our budget.

3. Select all the correct choices from the list below. 3 / 4 points
- ☐ $W(0, 1) = 0$

☒ $W(1, 1) = 2$

☒ Correct

Correct. This follows by applying the recurrence once and looking at the base cases.

☐ $W(1, 1) = \min(W(0, 1), W(1, 0) + 2)$

☒ $W(0, 0) = 0$

☒ Correct

Correct

You didn't select all the correct answers

